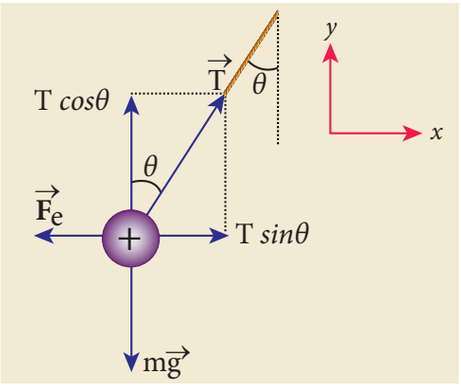
hanged vertically. Since they are positively charged spheres, there will be a repulsive force between them and they will be at equilibrium with each other at an angle of 30° with the vertical. At equilibrium, each charge experiences zero net force in each direction. We can draw a free body diagram for one of the charged spheres and apply Newton's second law for both vertical and horizontal directions.

The free body diagram is shown below.



In the x-direction, the acceleration of the charged sphere is zero.

Using Newton's second law $\left(\vec{F}_{tot} = m\vec{a}\right)$, we have

$$T\sin\theta\,\hat{i} - F_e\hat{i} = 0$$

$$T\sin\theta = F_e \qquad (1)$$

Here T is the tension acting on the charge due to the string and F_e is the electrostatic force between the two charges.

In the y-direction also, the net acceleration experienced by the charge is zero.

$$T\cos\theta\,\hat{j} - mg\,\hat{j} = 0$$

$$T\cos\theta = mg\ . \qquad (2)$$

By dividing equation (1) by equation (2),

$$\tan\theta = \frac{F_e}{mg} \qquad (3)$$

Since they are equally charged, the magnitude of the electrostatic force is

$$F_e = k\frac{q^2}{r^2} \quad \text{where } k = \frac{1}{4\pi\epsilon_0}$$

Here $r = 2a = 2L\sin\theta$. By substituting these values in equation (3),

$$\tan\theta = k\frac{q^2}{mg\left(2L\sin\theta\right)^2} \qquad (4)$$

Rearranging the equation (4) to get q

$$q = 2L\sin\theta\sqrt{\frac{mg\tan\theta}{k}}$$

$$= 2\times0.1\times\sin30^\circ\times\sqrt{\frac{10^{-3}\times10\times\tan30^\circ}{9\times10^9}}$$

$$q = 8.01\times10^{-8}\ \text{C} = 80.1\ \text{nC}$$

EXAMPLE 1.4

Calculate the electrostatic force and gravitational force between the proton and the electron in a hydrogen atom. They are separated by a distance of 5.3×10^{-11} m. The magnitude of charges on the electron and proton are 1.6×10^{-19} C. Mass of the electron is $m_e = 9.1\times10^{-31}$ kg and mass of proton is $m_p = 1.6\times10^{-27}$ kg.

Solution

The proton and the electron attract each other. The magnitude of the electrostatic force between these two particles is given by

$$F_e = \frac{ke^2}{r^2} = \frac{9\times10^9\times\left(1.6\times10^{-19}\right)^2}{\left(5.3\times10^{-11}\right)^2}$$

$$= \frac{9\times2.56}{28.09}\times10^{-7} = 8.2\times10^{-8}\ \text{N}$$

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